Provability facts for the conditional in tableaux

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-tableaux/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: all node formulas in this proof use the standard two-argument macro form; the trees render as genuine branching tableaux under prooftrees

Editorial note. The label prop:provability-lif names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Signed tableau rules for the conditional. From True (A implies B) the True-implies rule derives False A and True B on the branch; from False (A implies B) the False-implies rule branches into True A and False B.. Item (1) closes via the True-implies derivations against True A and False B; item (2) closes via the False-implies branches together with the True-not rule in the first tree.
- Signed tableau rules for negation. From True (not A) the True-not rule derives False A.. The first tree of item (2) closes its True A branch by applying the True-not rule to the assumed True (not A).

Proposition 0.1. *1.* $A, A \to B \vdash B$.

2. Both $\neg A \vdash A \to B$ *and* $B \vdash A \to B$.

Proof. 1. $\{\bot(B), \top(A \to B), \top(A)\}$ has a closed tableau:

1.	$\bot(B)$	Ass
2.	$\top(A \to B)$	Ass
3.	$\top(A)$	Ass
4.	$\bot(A)$ $\quad$ $\top(B)$ $\otimes$ $\quad$ $\otimes$	$\top\to$[2]

2. Both $\{\bot(A \to B), \top(\neg A)\}$ and $\{\bot(A \to B), \top(B)\}$ have closed tableaus:

1.	$\bot(A \to B)$	Ass
2.	$\top(\neg A)$	Ass
3.	$\top(A)$	$\bot\to$[1]
4.	$\bot(B)$	$\bot\to$[1]
5.	$\bot(A)$	$\top\neg$[2]
	$\otimes$	

1.	$\bot(A \to B)$	Ass
2.	$\top(B)$	Ass
3.	$\top(A)$	$\bot\to$[1]
4.	$\bot(B)$	$\bot\to$[1]
	$\otimes$	

□